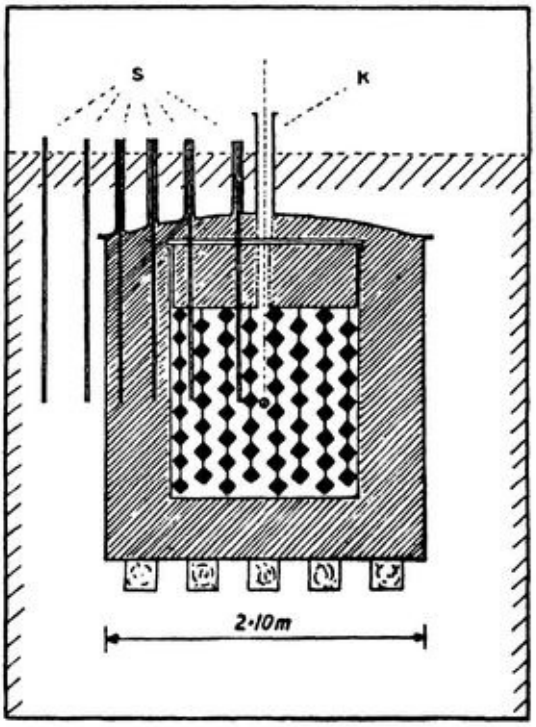



Figure 40B.—Plan of uranium pile.

The uranium reactor has a feature which makes it difficult to use it as a source of energy, but on the other hand, makes it very useful in another respect: It is filled by an extraordinarily intense radioactive radiation, which is extremely dangerous to every living being in the vicinity of the reactor. For this reason, the reactor is shielded by walls many metres in thickness, constructed of concrete or a like material. The reason for the presence of these radiations is obvious: The interior of the uranium reactor is the scene of nuclear reactions, occurring on a vast scale, which produce radiation of all kinds (alpha, beta and gamma radiation). An effect, in particular, of the gigantic neutron intensity in the reactor is that substances inserted into the apparatus very quickly become radioactive by the processes discussed in Section 4 of our Sixth Lecture. Thus the uranium reactor can be used as an especially efficient ‘crucible’ for artificially radioactivated substances. In fact, this has hitherto been the technically most important application of the uranium pile. For instance, plutonium, so important today as an ‘explosive’ for military purposes, is produced exclusively by a transmutation of ordinary uranium in the uranium reactor. This brings us next, after the question of the generation of energy, to the second main problem of nucleonics, which may be called ‘ennoblement of matter’.

## IV. ENNOBLEMENT OF MATTER BY NUCLEAR REACTIONS